Quarry: Technical Architecture Specification

Local Semantic Search for AI Agents

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1 Overview

Quarry is a local semantic search engine. It ingests documents in 20+ formats, embeds them with a local ONNX model, stores vectors in LanceDB, and serves semantic search to AI agents (Claude Code, Claude Desktop) and a macOS menu bar app.

The system follows the Punt Labs projection strategy: a single core library projects outward through four thin adapter layers. Each surface imports core functions and calls them with resolved arguments. No intermediary service layer.

1.1 Design Principles

P1. **Library-first.** The core library does all work. CLI, MCP, HTTP, and plugin surfaces are thin wrappers that resolve settings and delegate.

P2. **Local by default.** Everything runs on the user's machine. No API keys, no cloud accounts. The embedding model downloads once (∼120 MB, int8 quantized).

P3. **One daemon, many clients.** A single `quarry serve` process loads the embedding model once and serves all sessions over WebSocket. Clients connect via `mcp-proxy`.

P4. **Fire-and-forget I/O.** Side-effect MCP tools return immediately and process in a bounded thread pool. Search is synchronous.

P5. **Named databases.** Each database is fully isolated: its own LanceDB directory, sync registry, and vector index. Work/personal separation without configuration files.

2 System Architecture

2.1 Deployment Topology

```
                  stdio WebSocket
Claude Code <------------------> mcp-proxy <-------------------> quarry serve
            MCP JSON-RPC (~5 MB Go) (one daemon)

Claude Desktop <--- stdio ---> quarry mcp
                              (standalone process, no daemon)
```

Without the proxy, every Claude Code tab spawns a separate Python process, each loading the embedding model into ∼200 MB of RAM. With it, startup is instant and state is shared across all sessions.

The CLI (`quarry find`, `quarry ingest`, etc.) uses the library directly — no daemon involved. Each invocation loads the embedding model, does the work, and exits.

2.2 Surface Summary

Surface	Transport	Purpose
CLI	Direct library calls	Interactive use from terminal
MCP server	stdio or WebSocket	Claude Code/Desktop tools
HTTP server	REST JSON + WS `/mcp`	Menu bar app + mcp-proxy
Plugin	Hook shell commands	Auto-register, auto-ingest, compaction safety

3 Module Responsibilities

3.1 Data Layer

Module	File	Responsibility
Models	`src/quarry/models.py`	Immutable dataclasses: `PageType` enum, `PageContent`, `Chunk`, `PageAnalysis`. All `@dataclass(frozen=True)`.
Types	`src/quarry/types.py`	Protocol definitions for external dependencies: `LanceDB`, `LanceTable`, `LanceQuery`, `ListTablesResult`, `OcrBackend`, `EmbeddingBackend`. Structural typing instead of `Any`.
Config	`src/quarry/config.py`	`Settings` (pydantic `BaseSettings`), `resolve_db_paths()` for named databases, ONNX model constants.
Database	`src/quarry/database.py`	LanceDB operations: `insert_chunks`, `search`, `list_documents`, `delete_document`, `count_chunks`, etc. Single table `chunks`.

3.2 Ingestion Pipeline

Module	File	Responsibility
Pipeline	`src/quarry/pipeline.py`	Orchestrator: dispatches by format, chunks, embeds, stores. Entry points: `ingest_document`, `ingest_content`, `ingest_auto` (URL with sitemap discovery).
Chunker	`src/quarry/chunker.py`	Text splitting with overlap. Respects page boundaries and heading structure.
Collections	`src/quarry/collections.py`	`derive_collection()` — determines collection name from path or URL.

3.3 Format Processors

Each processor converts a source format into `list[PageContent]`:

Module	File	Formats
PDF	`src/quarry/pdf_analyzer.py`	PDF (text via PyMuPDF, OCR fallback for image pages)
Image	`src/quarry/image_analyzer.py`	PNG, JPG, TIFF, BMP, WebP (OCR)
Text	`src/quarry/text_processor.py`	TXT, MD, LaTeX, DOCX
Code	`src/quarry/code_processor.py`	30+ languages (AST-aware splitting)
HTML	`src/quarry/html_processor.py`	HTML (boilerplate stripping, Markdown conversion)
Spreadsheet	`src/quarry/spreadsheet_processor.py`	XLSX, CSV (tabular serialization)
Presentation	`src/quarry/presentation_processor.py`	PPTX (slide-per-chunk with notes)
Dispatcher	`src/quarry/text_extractor.py`	Extracts text from text-based PDF pages (PyMuPDF)

3.4 Backends

Module	File	Responsibility
Factory	`src/quarry/backends.py`	Thread-safe factory with double-checked locking. `get_ocr_backend()` and `get_embedding_backend()` return cached singletons.
OCR	`src/quarry/ocr_local.py`	Local OCR via RapidOCR (offline, no setup).
Embeddings	`src/quarry/embeddings.py`	Local ONNX embedding (snowflake-arctic-embed-m-v1.5, 768-dim, 512 tokens).

3.5 Sync and Registry

Module	File	Responsibility
Sync	`src/quarry/sync.py`	Directory sync: discover files, compute delta (new/changed/deleted), ingest/delete accordingly. `ThreadPoolExecutor` for concurrent ingestion.
Registry	`src/quarry/sync_registry.py`	SQLite registry of watched directories. Maps directories to collections with timestamps.

3.6 Surfaces

Module	File	Responsibility
CLI	`src/quarry/__main__.py`	Typer CLI with rich formatting. Synchronous. All commands: resolve settings → call core → format output.
MCP Server	`src/quarry/mcp_server.py`	FastMCP server. Read-only tools are synchronous. Side-effect tools use fire-and-forget.
HTTP Server	`src/quarry/http_server.py`	Starlette ASGI app on uvicorn. REST endpoints + WebSocket `/mcp`.
Hooks	`src/quarry/hooks.py`	Claude Code plugin hooks: session-start (auto-register), post-web-fetch (auto-ingest), pre-compact (capture transcript). All fail-open.

3.7 Support

Module	File	Responsibility
Formatting	`src/quarry/formatting.py`	Pre-formatted plain text output using unicode box-drawing. Used by both CLI and MCP for consistent output.
Results	`src/quarry/results.py`	`SyncResult` dataclass for sync operation reporting.
Sitemap	`src/quarry/sitemap.py`	Sitemap discovery and URL extraction for web ingestion.
Doctor	`src/quarry/doctor.py`	Health checks and install flow: model availability, database access, MCP configuration, ethos extension setup.

4 Daemon Model

4.1 Why a Daemon

The embedding model (snowflake-arctic-embed-m-v1.5, int8 quantized ONNX) occupies ∼200 MB of RAM and takes ∼250 ms to load. MCP's stdio transport model spawns a new process per session.

Without a daemon, five Claude Code tabs means five copies of the model in memory and five cold starts.

`quarry serve` loads the model once and keeps it resident. The `mcp-proxy` binary (∼5 MB Go binary, <10 ms startup) bridges each session's MCP stdio to the daemon over WebSocket.

4.2 mcp-proxy Architecture

`quarry install` downloads `mcp-proxy` (SHA256-verified, platform-specific binary) and configures MCP clients to use it. The proxy:

1. Reads MCP JSON-RPC from stdin.
2. Forwards to `ws://localhost:8420/mcp` over WebSocket.
3. Relays responses back to stdout.

Each WebSocket connection gets its own asyncio Task with a `ContextVar` for database selection (`_db_name`), so `use("work")` in one session does not affect others.

4.3 CLI Independence

The CLI does not use the daemon. `quarry find "margins"` loads the embedding model in-process, queries LanceDB, prints results, and exits. LanceDB handles concurrent reads safely, so CLI and daemon can access the same database simultaneously.

5 MCP Tool Surface

Tool	Key Parameters	Purpose	Exec
`find`	`query`, `limit`, `collection`	Semantic search	Sync
`ingest`	`source`, `overwrite`, `collection`	Ingest file or URL	Bg
`remember`	`content`, `document_name`	Ingest inline text	Bg
`list`	`kind`	List indexed data	Sync
`show`	`document_name`, `page_number`	Document metadata or page	Sync
`delete`	`name`, `kind`	Delete document or collection	Bg
`register_directory`	`directory`, `collection`	Register for sync	Bg
`deregister_directory`	`collection`, `keep_data`	Remove registration	Bg
`sync_all`	—	Sync all registered dirs	Bg
`status`	—	Database stats	Sync
`use`	`name`	Switch active database	Sync

Bg = background (`ThreadPoolExecutor`).

Background tools use a bounded `ThreadPoolExecutor(max_workers=4)`. They return an optimistic response immediately. Settings and database connections are snapshotted at the tool boundary and passed into background functions — this avoids a race condition where the mutable `_db_name` ContextVar could change between the tool call and the background thread's execution. Exceptions in background threads are logged via `logger.exception()`, not raised.

6 HTTP Server

6.1 ASGI Application

Starlette app built by `build_app()` factory, served by uvicorn. REST handlers are sync functions — Starlette auto-detects this and runs them in its threadpool. The `/mcp` WebSocket endpoint is async.

Default port: 8420 (`DEFAULT_PORT` in `src/quarry/config.py`), overridable with `--port`.

6.2 Endpoints

Path	Method	Purpose	Auth
/health	GET	{"status":"ok", "uptime_seconds":N}	No
/search	GET	Vector-only search (q, limit, collection, page_type, source_format)	Yes
/documents	GET	List indexed documents	Yes
/collections	GET	List collections	Yes
/status	GET	Database statistics	Yes
/mcp	WS	MCP JSON-RPC over WebSocket	Yes

6.3 Port File

After binding, the server writes the actual port to ~/.punt-labs/quarry/data/<db>/serve.port. This supports --port 0 (ephemeral assignment). Removed on shutdown via lifespan context.

7 Configuration

7.1 Settings

All settings are environment variables read via pydantic-settings.

Variable	Default	Description
Paths		
QUARRY_ROOT	~/.punt-labs/quarry/data	Base directory for all databases
LANCEDB_PATH	<root>/default/lancedb	Vector database location
REGISTRY_PATH	<root>/default/registry.db	Directory sync SQLite database
LOG_PATH	~/.punt-labs/quarry/logs/quarry.log	Log file (rotating, 5 MB, 3 backups; independent of QUARRY_ROOT)
Chunking		
CHUNK_MAX_CHARS	1800	Target max characters per chunk (∼450 tokens)
CHUNK_OVERLAP_CHARS	200	Overlap between consecutive chunks
Embedding		
EMBEDDING_MODEL	Snowflake/snowflake-arctic-embed-m-v1.5	Model identifier (display only — the ONNX model is fixed)
EMBEDDING_DIMENSION	768	Vector dimension (display only — fixed at 768)

7.2 Named Databases

resolve_db_paths(settings, name) maps a database name to a path under QUARRY_ROOT. Each database is fully isolated — its own LanceDB directory, sync registry, and vector index. Names containing path separators (/ or \) or the literal traversal segments . or .. are rejected; names like foo.bar are allowed. If LANCEDB_PATH was explicitly overridden, the override is preserved.

The persistent default database is stored in ~/.punt-labs/quarry/config.toml as [default] database = "name".

7.3 Chunking Tuning

Increase CHUNK_MAX_CHARS (3000–4000) for documents with long self-contained sections: legal contracts, academic papers, large source code functions.

Decrease `CHUNK_MAX_CHARS` (800–1200) for dense documents where each paragraph covers a distinct topic.

Increase `CHUNK_OVERLAP_CHARS` (400) when relevant information spans paragraph boundaries.

The embedding model has a 512-token context window. Chunks exceeding this are truncated during embedding, reducing quality for the tail.

8 Search and Retrieval

8.1 Embedding Model

snowflake-arctic-embed-m-v1.5 via ONNX Runtime:

- 768-dimensional vectors, 512-token context window
- Asymmetric retrieval: queries are prefixed with a search instruction, documents are not
- int8 quantized ONNX for CPU (∼120 MB), FP16 for CUDA (∼220 MB)
- Provider auto-detection: CUDA+FP16 when available, CPU+int8 fallback. `QUARRY_PROVIDER` env var overrides (`cpu`, `cuda`, or unset for auto). See DES-016 and `provider.py`.
- Auto-downloads on first use via `huggingface_hub`

The model is fixed — there is no configuration to swap it. Changing the model invalidates all existing embeddings, requiring full re-ingestion.

8.2 OCR

RapidOCR handles all image-based text extraction. For semantic search, OCR accuracy matters less than expected — the embedding model is robust to minor OCR errors. A misspelled word rarely changes semantic meaning enough to affect search ranking.

8.3 Search Quality Tips

1. **Use natural language queries.** The model is trained on question-passage pairs. "What were Q3 revenue figures?" works better than "Q3 revenue."
2. **Use collection filtering.** `quarry find "auth logic" -c backend`.
3. **Re-ingest after tuning.** Chunk parameters only affect new ingestions. Run `quarry sync` after changes.
4. **Check chunk boundaries.** `quarry find -n 1 "query"` and examine the result.
5. **Use `--overwrite` when re-ingesting.** Without it, duplicate chunks accumulate.

8.4 Hybrid Search

When an `agent_handle` filter is provided, quarry uses dual-channel retrieval instead of vector-only search.

Channels:

1. **Vector similarity** — existing ANN search via LanceDB
2. **BM25 full-text search** — Tantivy index on the `text` column, created during schema migration

Results are fused using Reciprocal Rank Fusion (RRF):

$$\text{score}(m) = \sum_{\text{channel}} \frac{1}{60 + \text{rank}_{\text{channel}}(m)}$$

Temporal decay applies to agent-scoped memories (non-empty `agent_handle`):

$$w = e^{-\lambda \cdot h}$$

where λ is the decay rate and h is hours since ingestion. Unscoped documents (`agent_handle` = "") are exempt from decay.

When no `agent_handle` is provided, search uses the existing vector-only path (backwards compatible).

8.5 Agent Memory Schema

Three columns extend the chunks table for agent-scoped memory:

Column	Type	Default	Purpose
`agent_handle`	utf8	`""`	Agent identity scope
`memory_type`	utf8	`""`	fact, observation, opinion, procedure
`summary`	utf8	`""`	One-line summary

Migration is idempotent — `_migrate_schema()` adds missing columns with empty-string defaults on every table open. Existing databases gain the columns transparently.

Identity tagging: the PreCompact hook reads the agent handle from `.punt-labs/ethos/config.yaml` and tags captured transcripts automatically.

8.6 Ethos Extension Integration

Quarry stores its configuration in ethos's generic extension mechanism (DES-008, DES-019):

```
~/.punt-labs/ethos/identities/<handle>.ext/quarry.yaml
```

The extension file contains two categories of data:

1. **Sidecar config** — `memory_collection` and `expertise_collections` are read directly by quarry via the filesystem (no ethos API call).
2. **Session context** — `session_context` is a YAML literal block scalar emitted verbatim by ethos at session start and before context compaction. It contains the agent's memory instructions (collection name, slash commands, memory types).

`quarry install` (step 7/7) scans all identity extension directories and appends `session_context` to any `quarry.yaml` that has `memory_collection` but no `session_context`. Uses raw file append to preserve existing YAML comments and formatting. Per-identity exception handling ensures one malformed file does not abort the rest.

Ownership boundary: ethos emits quarry's instructions without understanding them. Quarry writes its own instructions without importing ethos. The dependency is one-way — quarry depends on ethos for identity; ethos has zero knowledge of quarry's internals.

9 Logging and Exceptions

9.1 Logging Standard

Every module that performs I/O declares `logger = logging.getLogger(__name__)`. Pure data modules (`models.py`, `types.py`, `chunker.py`) do not.

Logging is configured once at the application entry point (CLI or MCP server). Library modules never call `logging.basicConfig`.

Level	Use
`DEBUG`	Internal state: variable values, branch decisions, intermediate results.
`INFO`	Operational milestones: starting a job, completing a step, resource counts.
`WARNING`	Unexpected but recoverable: empty input, deprecated parameter usage.
`ERROR`	Failures that prevent completion but do not crash. Always with exception context.

`CRITICAL` is not used. If the process must exit, it raises an exception.

Log messages use `%s`-style formatting (lazy evaluation), not f-strings. Each message includes enough context to trace an operation without reading code.

9.2 Exception Handling

The codebase uses built-in exception types. No custom exceptions unless a caller needs to distinguish failure modes programmatically.

Exception	Raised when
`ValueError`	Invalid input: unsupported format, bad parameter.
`FileNotFoundError`	A file path argument does not exist.
`RuntimeError`	An operation reports failure (OCR engine error, ingestion failure).
`TimeoutError`	An external service exceeds its time budget.

Rules:

1. Never swallow exceptions. Every `except` block must re-raise, or log with `logger.exception()` and raise a different exception, or (at system boundaries only) return a documented sentinel.
2. Never use bare `except:`.
3. Never catch `Exception` broadly in library code. Broad catches are permitted only at the outermost boundary (MCP tool handler, CLI command).
4. Exceptions carry context: include the values that caused the failure.
5. Use `try/finally` for cleanup, not `try/except`.
6. Do not use exceptions for flow control.

10 Security

10.1 Bearer Token Authentication

`quarry serve --api-key` (or `QUARRY_API_KEY` env var) gates all HTTP endpoints except `/health` behind `Authorization: Bearer <key>`.

- `/health` is exempt — load balancers need unauthenticated health checks.
- `OPTIONS` is exempt — CORS preflight requests carry no auth.
- `hmac.compare_digest` for token comparison — prevents timing attacks.
- Case-insensitive scheme per RFC 7235.
- Empty key = no auth — prevents accidental trivial bypass.
- No key = auth disabled — local use requires no key.

The server refuses to bind to a non-loopback address without `--api-key`.

10.2 WebSocket Security (CSWSH)

Origin-based cross-site WebSocket hijacking protection:

- If an `Origin` header is present, it must match the CORS allow-list.
- Browsers always send `Origin`; non-browser clients like `mcp-proxy` do not.
- Bearer auth is checked before WebSocket accept (close code 1008 on failure).

10.3 CORS

`CORSMiddleware` configured with GET and OPTIONS methods, Authorization and Content-Type headers. Default allowed origin: `http://localhost`. Configurable via `--cors-origin` (repeatable).

The server reflects the request's `Origin` header only when it matches the allow list, and adds `Vary: Origin` to signal cacheability varies by origin. When no `Origin` is present, no CORS headers are emitted.

10.4 Log Safety (CWE-532, CWE-117)

Uvicorn's access log is disabled (`access_log=False`), eliminating query string leakage at the source. The search handler logs only result count, never the raw query. WebSocket session keys are sanitized before logging: control characters stripped via `_CONTROL_CHAR_RE` and truncated to 64 characters to prevent log injection (CWE-117) and log flooding.

11 Deployment

11.1 Local (CLI)

```
uv tool install punt-quarry
quarry install # downloads model + mcp-proxy
quarry doctor # verifies environment
```

11.2 Daemon (`quarry serve`)

```
quarry serve --port 8420 --api-key $QUARRY_API_KEY
```

Managed by `launchd` (macOS) or `systemd` (Linux) via `quarry install`.

11.3 Container (Fly.io)

Multi-stage Dockerfile, three stages:

1. **deps** (`python:3.13-slim`): installs `uv`, runs `uv sync --frozen --no-dev`. Two-pass for Docker layer caching (deps first, then source).
2. **model** (extends `deps`): downloads the embedding model at build time (∼120 MB, int8 quantized). Cached as a separate Docker layer.
3. **runtime** (`python:3.13-slim`): copies `/app` from deps, copies model cache from model stage. `QUARRY_ROOT=/data`, port 8080.

LanceDB data lives on a Fly persistent volume at `/data`. `QUARRY_API_KEY` is set via `fly secrets`. Cold start is ∼5 s with the baked-in model.

12 Performance

12.1 Critical Path

Operation	Latency	Notes
Model load	∼250 ms	One-time per process
Embedding (1 chunk)	∼50 ms	Apple Silicon, CPU
LanceDB search	<10 ms	Vector similarity + filter
End-to-end search	<100 ms	After model is loaded

12.2 Fire-and-Forget MCP Pattern

Side-effect MCP tools return immediately and process in a bounded thread pool (4 workers). This is necessary because:

1. MCP tool calls block the LLM's response stream. A 30-second ingest blocks Claude from responding.
2. The bounded pool prevents resource exhaustion under concurrent requests.
3. Settings and database connections are snapshotted at the tool boundary, avoiding race conditions with the mutable `_db_name` ContextVar.

12.3 Single Table Design

All chunks live in one LanceDB table (`chunks`). Document and collection boundaries are columns, not separate tables. This simplifies cross-document search and avoids table proliferation. Filtering by document/collection/page_type/source_format uses LanceDB's built-in filter predicates on the vector search.

13 Plugin and Hook System

The Claude Code plugin (`.claude-plugin/`) registers hooks that fire on session events and tool calls. All hooks are fail-open: exceptions are logged and the process exits 0.

Hook Event	Matcher	Effect	Blocks?
SessionStart	*	Auto-register directory + background sync	No
PostToolUse	quarry tools	Suppress tool output formatting	No
PostToolUse	WebFetch	Auto-ingest fetched URL	No
PreCompact	*	Capture transcript as session notes	No

13.1 Session Start

`handle_session_start()` auto-registers the working directory and launches a background sync via a detached subprocess. Uses atomic PID-file locking (`O_CREAT|O_EXCL`) at `~/.punt-labs/quarry/sync.pid` to prevent concurrent syncs. Returns `additionalContext` immediately without waiting.

13.2 Web Fetch Capture

`handle_post_web_fetch()` ingests URLs fetched during a session into the `web-captures` collection. Deduplicates by document name. Prefers already-fetched content from the tool response (avoids re-fetch, reduces SSRF surface).

13.3 Pre-Compact Capture

`handle_pre_compact()` reads the transcript (JSONL) and ingests it into the `session-notes` collection. Extracts user/assistant message text, skipping tool-use blocks. Capped at 500,000 characters.

14 CLI Commands

Command	Purpose
`ingest <source>`	Ingest file or URL. Flags: `--agent-handle`, `--memory-type`, `--summary`
`remember`	Ingest stdin content
`find <query>`	Semantic search
`show <document>`	Document metadata or page text
`status`	Database statistics
`use <name>`	Set persistent default database
`delete <name>`	Delete document or collection
`register <directory>`	Register for incremental sync
`deregister <collection>`	Remove directory registration
`sync`	Sync all registered directories
`list documents`	List indexed documents
`list collections`	List collections with counts
`list databases`	List named databases
`list registrations`	List registered directories
`serve`	Start HTTP API server

Command	Purpose
`mcp`	Start MCP server (stdio)
`install`	Set up data directory, download model, configure ethos ext
`doctor`	Check environment health
`uninstall`	Remove system daemon
`version`	Print version

Global flags: `--json`, `--verbose/-v`, `--quiet/-q`, `--db <name>`.

15 Test Architecture

821 tests across 33 files. Three tiers plus integration:

Tier	Strategy	Files
Tier 1: Core logic	Real data, no mocks	`test_database`, `test_chunker`, `test_embeddings`, `test_code_processor`, `test_html_processor`, `test_text_processor`, `test_collections`, `test_config`, `test_formatting`, `test_image_analyzer`, `test_latex_utils`, `test_models`, `test_pdf_analyzer`, `test_presentation_processor`, `test_spreadsheet_processor`, `test_text_extractor`, `test_registry` (17 files)
Tier 2: Orchestration	Mocked I/O	`test_pipeline`, `test_pipeline_images`, `test_ocr_local`, `test_url_ingestion`, `test_sitemap`, `test_backends`, `test_sync`, `test_hooks`, `test_doctor` (9 files)
Tier 3: Surface wiring	Fully mocked, <2 s	`test_cli`, `test_mcp_server`, `test_mcp_websocket`, `test_http_server`, `test_proxy`, `test_service` (6 files)
Integration	Real ONNX + real LanceDB	`test_integration` (`@pytest.mark.slow`)

Quality gates: `ruff` (lint + format), `mypy` (strict), `pyright` (strict), `pytest` (all markers except slow/integration). Zero violations required.

16 Formal Model

The plugin state machine — how quarry hooks into the Claude Code session lifecycle — is formally specified in `docs/claude-code-quarry.tex` using Z notation (ISO 13568).

The specification covers:

- Session start with/without auto-sync

- Web fetch auto-capture into `web-captures` collection
- Pre-compaction transcript capture into `session-notes`
- Database switching via `use` tool
- Knowledge hint injection at tool boundaries
- Compaction observability: a capture flag records whether capture preceded compaction; the hook system enforces capture-before-compaction on a best-effort basis
- Clean session boundaries: all quarry state cleared on `EndSession`

The specification can be type-checked with `fuzz -t claude-code-quarry.tex`.